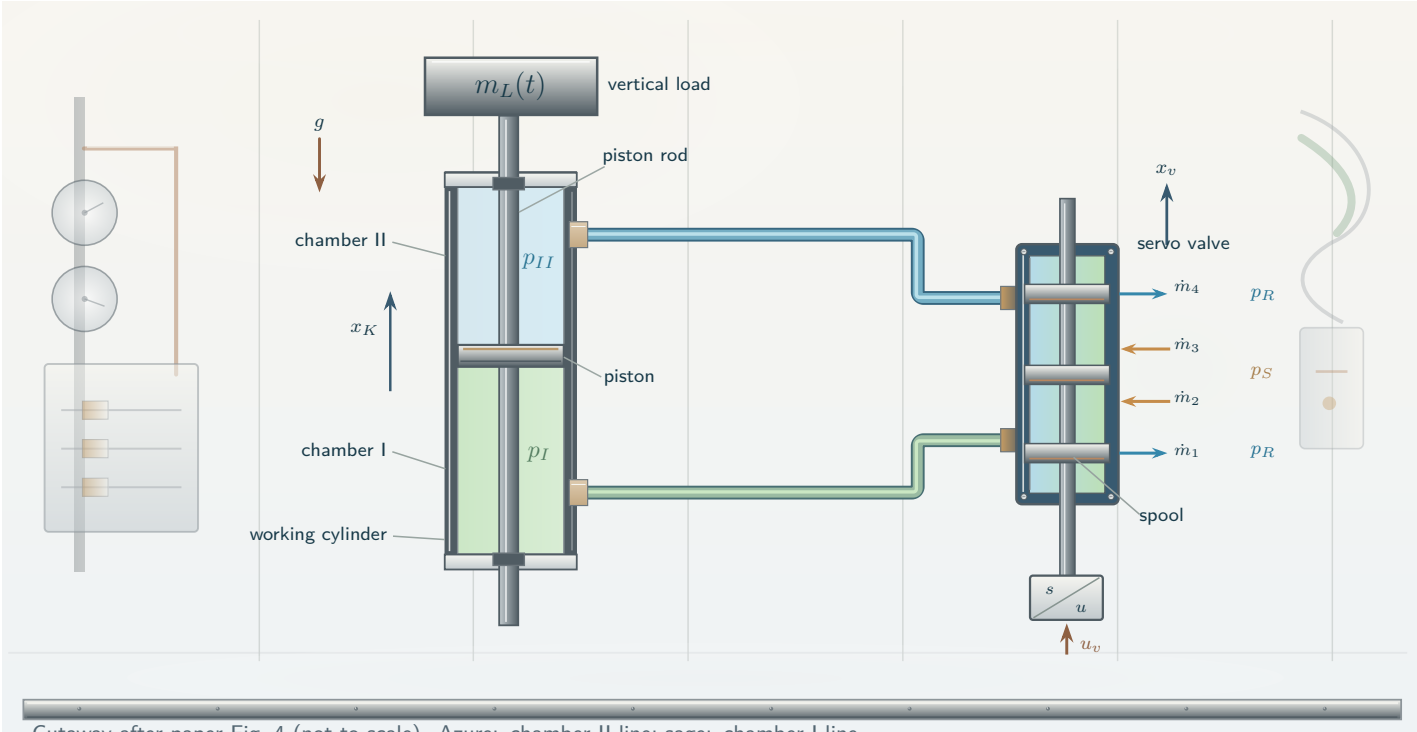
Vertical servo-pneumatic actuator

Paper section 3.5: *Servo-pneumatic actuator*

Double-rod cylinder, time-varying payload and servo-valve mass-flow paths



Cutaway after paper Fig. 4 (not to scale). Azure: chamber II line; sage: chamber I line.
Blue arrows: return; amber arrows: supply. The $\dot{m}_i$ signs may reverse.

Paper topology and inputs

$m_L(t)$: vertical load mass.
x_K: piston displacement.
x_v: spool displacement.
u_v: servo-valve command.

Four port rates $\dot{m}_{1\ldots4}$; supply p_S and return p_R.

Source and modeling scope

Figure topology: Schulte–Hahn (2004), cited in Kroll–Schulte (2014).

Local identified models use operating-point offsets. The SI conservation model is a separately parameterized teaching variant, not a calibrated replica.

Chamber volumes and pressures

$V_I = V_0 + Ax_K, \quad V_{II} = V_0 - Ax_K.$
$p_j = m_j RT_j / V_j.$
The double-ended rod supports equal effective piston areas in the model.

Vertical force (constant-payload variant)

$(M+m)\dot{v} = A(p_I - p_{II}) - f(v)$
$\quad -(M+m)g - F_{ext}.$
A changing $m_L(t)$ would require an additional momentum convention.

What is and is not identified

Literal survey equations (33)–(34) remain separate from local fitted models and the physical SI variant. Valve and heat-flow parameters require calibration.

Scientific cutaway based on the cited Fig. 4, not construction-ready CAD. Port arrows represent a sign convention; the published time-varying payload and the repository's constant-payload physical simulations require different assumptions. See PNEUMATIC_AUDIT.md.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.